Project Documentation format

1. Introduction

Project Title

An Employee Requests Installation of Licensed Software through the Service Catalog

Team Members

- Team Member 1: ServiceNow Developer
- Team Member 2: ServiceNow Administrator
- Team Member 3: Testing / Documentation
- Team Member 4: Workflow and Automation

Technology

- ServiceNow Platform
- Service Catalog
- Flow Designer
- ServiceNow Tables
- Business Rules
- Client Scripts
- UI Policies
- Notifications
- Roles and ACLs
- Reports and Dashboard

2. Project Overview

Purpose

The purpose of this project is to provide employees with a centralized and automated way to request licensed software through the ServiceNow Service Catalog.

The system eliminates manual communication through emails or separate requests by providing a structured process for submitting, approving, fulfilling, and tracking software installation requests.

Objectives

- Provide a simple Service Catalog item for software requests.
- Capture all required employee and software information.

- Automate the approval process.
- Route approved requests to the appropriate IT team.
- Verify software license availability.
- Create fulfillment tasks for IT support.
- Provide request status tracking.
- Send automatic notifications to employees.
- Maintain request and approval history.
- Improve transparency and reduce manual effort.

Key Features

- Licensed Software Service Catalog item.
- Software selection.
- Employee and device information.
- Business justification.
- Mandatory field validation.
- Automated request creation.
- Manager approval/rejection.
- Automated IT assignment.
- License availability verification.
- Software installation task.
- Request status tracking.
- Email notifications.
- Request closure.
- Reports and dashboards.
- Role-based access control

3. Architecture

3.1 ServiceNow Architecture

The application is developed on the ServiceNow platform using Service Catalog and workflow automation capabilities.

Main Flow

Employee → Service Catalog → Software Request → Manager Approval → IT Fulfillment → Software Installation → Notification → Request Closure

Employee Layer

- Employee accesses the ServiceNow portal.
- Employee opens the Service Catalog.
- Employee selects the licensed software request.
- Employee enters the required information.

- Employee submits the request.

Application / Workflow Layer

- ServiceNow creates the request and RITM.
- Flow Designer starts the approval workflow.
- Request is routed to the appropriate manager.
- Manager approves or rejects the request.
- Approved requests are routed to IT support.
- Fulfillment task is created.

Fulfillment Layer

- IT support checks license availability.
- IT support installs the software.
- Installation status is updated.
- Request is completed and closed.

Data Layer

ServiceNow stores information related to:

- Requested software
- Employee details
- Request details
- Approval information
- Fulfillment tasks
- Work notes
- Request status
- Assignment information
- Request history

4. Setup Instructions

Prerequisites

- Access to a ServiceNow instance.
- Appropriate ServiceNow user/admin role.
- ServiceNow Service Catalog access.
- Flow Designer access.
- System configuration access.
- Basic knowledge of ServiceNow administration.
- Required roles for creating catalog items and workflows.

Configuration Steps

1. Log in to the ServiceNow instance.
2. Open Service Catalog.
3. Create a new Catalog Item.
4. Provide the name Licensed Software Request.
5. Create the required catalog variables.
6. Configure mandatory fields.
7. Configure user criteria and access.
8. Create the approval flow using Flow Designer.
9. Configure manager approval.
10. Configure approval/rejection conditions.
11. Create IT fulfillment tasks.
12. Configure assignment groups.
13. Configure email notifications.
14. Configure request status updates.
15. Test the complete workflow.

5. Application Structure

Since this is a ServiceNow application, the structure is different from a MERN project.

Service Catalog

- Licensed Software Request
- Catalog Variables
- Variable Sets
- Catalog UI Policies
- Catalog Client Scripts
- User Criteria

Workflow / Automation

- Flow Designer
- Approval Actions
- Assignment Actions
- Notification Actions
- Request Status Updates

Data

- Request
- Requested Item (RITM)
- Catalog Task
- Approval records
- User information
- Assignment information

Administration

- Roles
- Groups
- ACLs
- Notifications
- Reports
- Dashboards

6. Running / Using the Application

The application is accessed through the ServiceNow portal rather than running commands such as `npm start`.

Employee Process

1. Log in to ServiceNow.
2. Open the Service Catalog.
3. Search for Licensed Software Request.
4. Open the catalog item.
5. Select the required software.
6. Enter employee/device information.
7. Enter business justification.
8. Submit the request.
9. Note the generated request/RITM number.
10. Track the request status.

Manager Process

1. Manager receives an approval request.
2. Reviews the requested software.
3. Reviews business justification.
4. Approves or rejects the request.
5. Adds comments if required.

IT Support Process

1. IT team receives the approved request.
2. Checks license availability.
3. Performs software installation.
4. Updates work notes.
5. Updates the task status.
6. Completes the request.

7. Workflow / Automation Documentation

The workflow automates the complete software installation process.

Workflow Steps

Step 1: Employee submits software request.

Step 2: ServiceNow creates the request/RITM.

Step 3: System validates the submitted information.

Step 4: Request is automatically sent to the manager.

Step 5: Manager approves or rejects the request.

Step 6: If rejected, the employee receives a rejection notification.

Step 7: If approved, the request is assigned to IT support.

Step 8: IT team verifies license availability.

Step 9: IT team installs the software.

Step 10: IT team updates the installation status.

Step 11: Employee receives completion notification.

Step 12: Request is closed.

8. Authentication and Authorization

ServiceNow provides authentication and role-based authorization.

Authentication

- Users log in using authorized ServiceNow credentials.
- Only authenticated users can access the application.
- User identity is used to determine access to requests.

Authorization

Different users have different responsibilities:

Employee

- Submit software requests.
- View their own requests.
- Track request status.

Manager

- Review requests.
- Approve or reject requests.
- Add approval comments.

IT Support

- View assigned requests.
- Check license availability.
- Perform installation.
- Update fulfillment tasks.

Administrator

- Configure catalog items.
- Manage workflows.
- Configure roles and access.
- Manage reports and system settings.

9. User Interface

The main user interface is the ServiceNow Service Portal / Employee Center.

Screenshots to Include

You can add screenshots of:

1. ServiceNow Login Page
2. Service Catalog
3. Licensed Software Catalog Item
4. Software Selection Field
5. Employee Information Form
6. Business Justification Field
7. Submitted Request
8. RITM Details
9. Manager Approval Screen
10. IT Fulfillment Task
11. Request Status
12. Notification

13. Completed Request
14. ServiceNow Report/Dashboard

10. Testing

Testing ensures that the software request process works correctly from submission to completion.

Functional Testing

- Verify that the catalog item is visible to authorized employees.
- Verify that required fields are mandatory.
- Verify that software can be selected.
- Verify that requests can be submitted.
- Verify that an RITM is generated.
- Verify that approval is sent to the correct manager.
- Verify approve/reject functionality.
- Verify IT task creation.
- Verify assignment to the correct IT group.
- Verify request status changes.
- Verify request closure.

Performance Testing

- Check Service Catalog loading time.
- Check request submission response time.
- Verify workflow execution time.
- Test multiple requests being submitted.
- Verify that the system handles concurrent users efficiently.
- Check notification processing.
- Verify that request tracking responds properly.

Security Testing

- Verify unauthorized users cannot access requests.
- Verify employees can view only permitted requests.
- Verify only managers can approve requests.
- Verify only authorized IT users can fulfill requests.
- Verify role-based access controls.
- Verify sensitive information is protected.

User Acceptance Testing

- Employee successfully submits a request.
- Manager successfully approves/rejects it.
- IT team successfully fulfills the request.

- Employee successfully tracks the request.
- Notifications are received correctly.
- Request is successfully completed and closed.

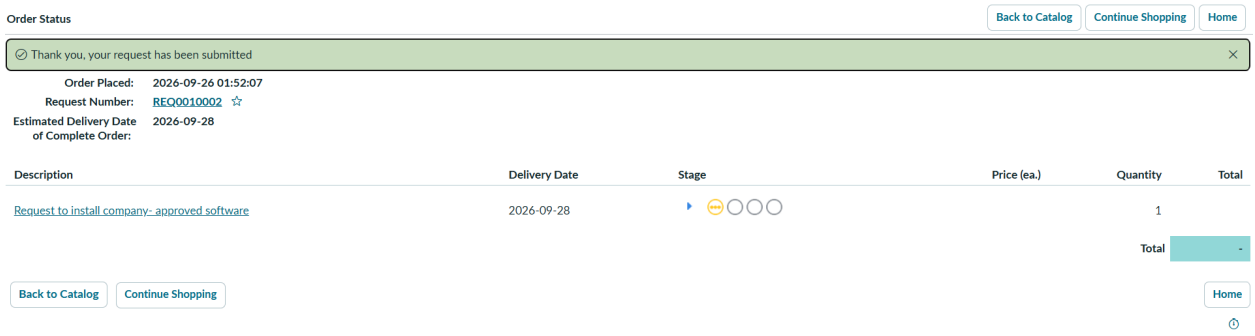
11. Screenshots / Demo

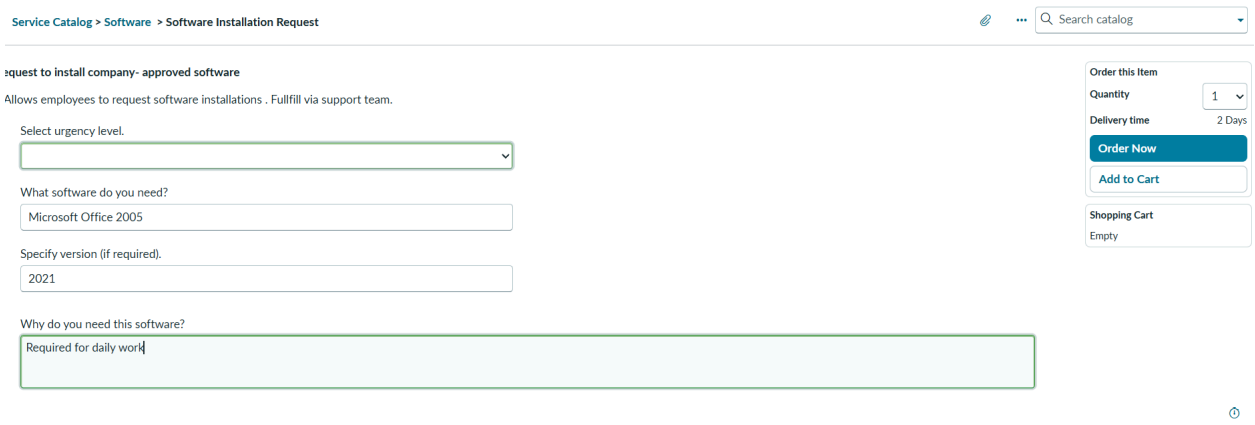
The project demonstration should show the complete workflow:

Login → Service Catalog → Licensed Software Request → Submit → Approval → IT Task → Installation → Notification → Closure

Screenshots

- Catalog Item
- Request Form
- Request Number
- Approval
- IT Task
- Status Tracking
- Notification
- Completed Request
- Dashboard/Report





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12. Known Issues

Possible limitations of the initial implementation:

- Software license availability may need to be maintained manually.
- Actual software installation may depend on the organization's endpoint management system.
- Approval time depends on manager availability.
- Additional software may require additional catalog configuration.
- External integrations may require separate authentication and configuration.
- Notification delivery may depend on ServiceNow email configuration.

13. Future Enhancements

1. AI / GenAI Integration — Optional

A GenAI feature can be added to analyze the employee's business justification and provide recommendations for the appropriate software.

2. Automated License Management

Integrate with a license management system to automatically check available licenses.

3. Automated Software Installation

Integrate ServiceNow with endpoint management tools so that approved software can be installed automatically.

4. Advanced Dashboard

Create dashboards showing:

- Total requests
- Pending requests
- Approved requests
- Rejected requests
- Completed requests
- Most requested software
- Average fulfillment time

5. Predictive Analytics

Historical request data can be analyzed to identify frequently requested software and improve license planning.

6. Mobile Access

Provide employees with easier access to software requests through mobile capabilities.

7. Automated Reminders

Send reminders to managers when approval requests remain pending for a specified period.

8. SLA Monitoring

Configure SLAs to monitor how quickly software requests are approved and fulfilled.

9. Knowledge Integration

Provide employees with relevant software installation guides and FAQs before submitting a request.

10. Additional Approval Rules

Different approval levels can be configured based on:

- Software cost
- Software category
- Department
- License type
- Security classification